

Oct. 31, 2021

My last three articles dealt with an interesting strategy found on the web. They are replicated here. They show one could use the constituent stocks of the QQQ ETF to build a prosperous long-term portfolio. All 3 articles cover the strategy's background, structure, potential, constraints, and limitations. Enclosed examples are intended to show, not only the feasibility, but also how easy it can all be. You will have the code, what it does, and what can be achieved. It will be up to you to apply this or not. I hope you will find it sufficiently interesting to also put it into good use.

Oct. 9, 2021

A Trading Strategy of Interest - Part I

Recently on QuantConnect.com, a trading strategy dealing with the QQQ ETF was published. You can find it [HERE](#). You can make a copy (clone it) and test it yourself should you want to.

The strategy caught my interest. All it did was rebalance QQQ's constituent stocks on a weekly basis. That was it. It represented an opportunity to study the trade mechanics of a pure-play rebalancing strategy in motion, something I wanted to revisit for some time.

QQQ is composed of the top 100 NASDAQ stocks by market cap. What could be the interest when you could just have bought QQQ and held it over the same time interval? Higher returns? Lower drawdowns? Something to do?

It might sound trivial. But QQQ is just an index tracker. It is designed to generate returns that try to mimic the NASDAQ 100 index. Why should continuously rebalancing the QQQ's constituents be better than buying it outright and holding it over the same time period?

The first thing to realize is that QQQ is by its very structure a market-cap-weighted average of its 100 stocks. Implying that individual price variations, both on the positive and negative side, could be much larger than QQQ's own price variations since QQQ is averaging things out.

The largest companies weigh in the most. The 10 largest stocks account for 55% of the total. The other 90 stocks make up the remaining 45% of this tracker. QQQ is definitely biased toward the highest valued stocks. It has also a strong bias toward technology: 45%. Note that those numbers change with time, just as the constituent stocks, but you get the idea, and you knew all that.

Holding on to QQQ for those 12.16 years would have produced a 21.37% CAGR due to QQQ's rise from 35.13 to 370.16 over the period. And, therefore, holding QQQ all by itself would not have been that bad an idea. At least, it would have done much better than having your money in a bank account earning low interests, or in bonds, or in some other fund averaging close to the secular market trend of about 10% or so.

The difference in outcomes can be considerable. Based on your initial trading capital (F_0), a 2% return would have given:

$$F(t) = F_0 \cdot (1 + 0.02)^{12.16} = 1.27 \cdot F_0$$

While a 10% rate would have produced:

$$F(t) = F_0 \cdot (1 + 0.10)^{12.16} = 3.18 \cdot F_0$$

The 21.37% QQQ CAGR would have generated:

$$F(t) = F_0 \cdot (1 + 0.2137)^{12.16} = 10.05 \cdot F_0$$

or 3.3 times more than the 10% scenario. This is like turning \$100,000 into \$1,005,000.

There was an incentive to buy and hold QQQ for the duration. A single investment decision and you would have beaten market averages just by buying this index surrogate, a simple index tracker.

The above-mentioned strategy, over the same period, generated a 19.01% CAGR. Was this reasonable? You have a tracker tracking an index tracker, what should you expect? Your portfolio should have about the same weighted distribution as QQQ since it was rebalanced every week to make sure it did. Those portfolio weights do not change that fast.

What I found interesting in this strategy was the ability to study the impact of rebalancing alone.

The strategy's 100k scenario made 28,856 trades vs the 1 trade if having bought QQQ outright and held. For the strategy, you had to monitor weekly what it did, put time and resources for the just in case something went wrong. While, buying QQQ was a one-shot decision, no need for any computer or software. You just bought and went along for the ride for those 12.16 years.

This is reminiscent of Mr. Buffett's million-dollar bet (which he won BTW). The bet then was that an active fund manager would not beat the S&P500 over the next 10 years. There was only one taker, and he lost. No other money manager was confident enough in his/her trading skills to take on the bet and make what should have been considered easy money. In itself, it does say something...

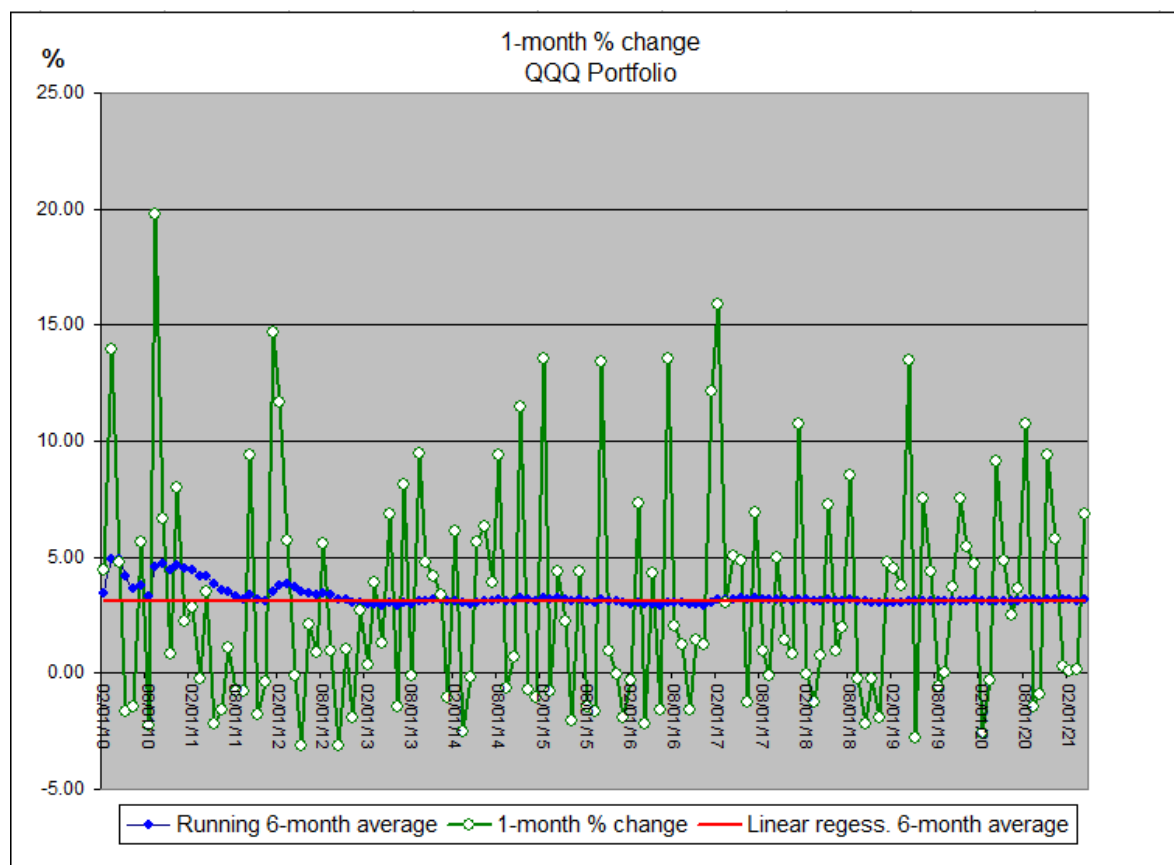
Most of the trading, except the initial purchases and the occasional stock change, were partial trades, small adjustments to the weights upon rebalancing.

Let's reduce the total trades by 200 to account for the 100 initial trades and for the removal and addition of stocks to the list. We are left with 28,656 partial trades due to the rebalancing procedure alone. However, a partial trade could occur only if the price variation was sufficient to buy or sell at least one share. Otherwise, no trade.

For example, a portfolio with 100k as starting capital with equal initial weights of 0.01, a single position in a 100-dollar stock would give you 10 shares. To buy or sell one more share would require the price to move by at least 10%. Often, even if the weights changed, it would not be sufficient to execute the weight reset. As it turned out, on average, some 45.27 partial trades per week were executed.

You had 28,656 trades that were executed with no trading logic other than the rebalancing procedure ($633 \cdot 45.27 = 28,656$). All these trades were not done on purpose meaning that they were planned in any way. They were just executed because the stock prices somehow changed for whatever reason. The strategy did not predict what was going to happen, it only reacted to the price change.

QQQ Monthly Percent Change



The chart above shows the monthly return for this trading strategy. As can be observed, it is rather erratic and could be considered short-term

unpredictable, saying that month-to-month predictions would be far from predictive. The red line is the linear regression over the total period. It could be known only after all the data was in. Its correlation coefficient R^2 is about 0.00001, saying that there is practically zero correlation over time. Nonetheless, there is a slight positive upside bias of $0.0004t$, again something saying that the curve is almost flat. The red line, nonetheless, is an expression for the average long-term upward bias.

The blue line represents the cumulative running 6-month average of those monthly returns. It can be observed that the blue line tends to the red line rather quickly (within 3 years) and then hugs the red line for most of the duration. This would hint at the notion that after some accumulated data we could have a relatively clear image of where it is all going.

The QQQ, by its very composition, is a bet on America all by itself. Its constituents are the 100 most prosperous and most highly valued stocks out there. The cream of what the market has to offer. A lot of wise choices are made for you by the QQQ's changing composition. Stocks that cannot keep pace with the best are removed while newcomers are added to the existing mix. No one is asking your advice on this, it is just internally executed. There is no second-guessing, your input is not requested or even considered in any way.

$$F(t) = F_0 + \sum (H_{QQQ} \cdot \Delta P) \rightarrow F_0 + \sum (h_Q \cdot \Delta p_Q)$$

The above equation is trying to convey that trading the 100 stocks in the QQQ ETF should tend to the same result as trading QQQ alone. Here, H_{QQQ} , is the ongoing holding matrix for the 100 stocks, whereas, h_Q is only holding QQQ. This outcome should be implicit. You are tracking an index tracker after all. The strategy's return should tend to QQQ's overall return.

Should you buy and hold DIA instead of QQQ, you would get a 14.15% return. While rebalancing the DIA's constituents would generate something like a 13.83% CAGR over the same interval. Holding DIA for the duration would have generated \$499,763, less than half of QQQ.

Again saying that it would be better to buy and hold DIA than rebalance

every week and execute some 13,497 trades.

One should not be surprised if the equation below also holds.

$$F(t) = F_0 + \sum(H_{DIA} \cdot \Delta P) \rightarrow F_0 + \sum(h_D \cdot \Delta p_D)$$

There would appear to be no strict advantage in tracking an index tracker. Technically, that is what this strategy's pure-play rebalancing demonstrated. Rebalancing alone, without any other trading logic, might not be sufficient to outperform an index tracker ETF.

Buying DIA or QQQ was just one decision. Both are index trackers. Both deal with some of the most valuable stocks out there. Both representing Mr. Buffett's bet on America, and yet, simply selecting QQQ over DIA for the duration would have generated twice as much.

Also, this might mark what a trading strategy should do as a bare minimum before being considered. If a strategy cannot exceed QQQ's overall return, its intrinsic value should seriously be put in doubt.

Oct. 20, 2021

A Trading Strategy of Interest - Part II

More observations. To put this in context, the last section presented a 12.16-year trading strategy simulation on a pure rebalancing play using the 100 stocks in QQQ, a weighted by market capitalization ETF. The strategy generated a 19.01% CAGR over the period, turning 100k into 830k or 1 Mil into 10.0 Mil with a 20.6% CAGR. Something better than most long-term market averages. Where to get a free copy of the program was also provided

Rebalancing QQQ on a weekly basis maintained the portfolio fully invested for the duration. All changes in the stock composition of QQQ were followed as they occurred. This led to mimicking QQQ's performance over the trading interval, or at least, getting close to it, return-wise. It is understandable, you have a 1:1 weighted relationship

with each of the 100 stocks in QQQ. Rebalancing was tracking QQQ's weights which, in itself, is an index tracker of a market-cap weighted index (NASDAQ 100). The performed simulations demonstrated quite well the side effects of this pure rebalancing play.

The number of shares to buy (q) for any trade (i) on any given day (d) for any given stock (j) has a floored value function: $\lfloor q_{i,d,j} \rfloor$.

For all trades, the position value was divided by the then-current price to obtain the number of shares to trade $\frac{w_{i,d,j} \cdot F(t)_{d-1}}{p_{i,d,j}}$. It was then floored to remove the decimal part.

$$\lfloor q_{i,d,j} \rfloor = \frac{F(t)_{d-1}}{J \cdot p_{i,d,j}} + \epsilon = \frac{w_{i,d,j} \cdot F(t)_{d-1}}{p_{i,d,j}} + \epsilon \quad \text{with} \quad w_{i,d,j} = \frac{1}{J}$$

Weight differences were marginalized and relatable to the flooring of share numbers. For example, no 122.75 shares, only integers allowed, and therefore, 122 shares traded, thereby tracking only slightly below QQQ's weights. The flooring truncation, as small as it might seem, does impact the non-trading zone described in the previous section.

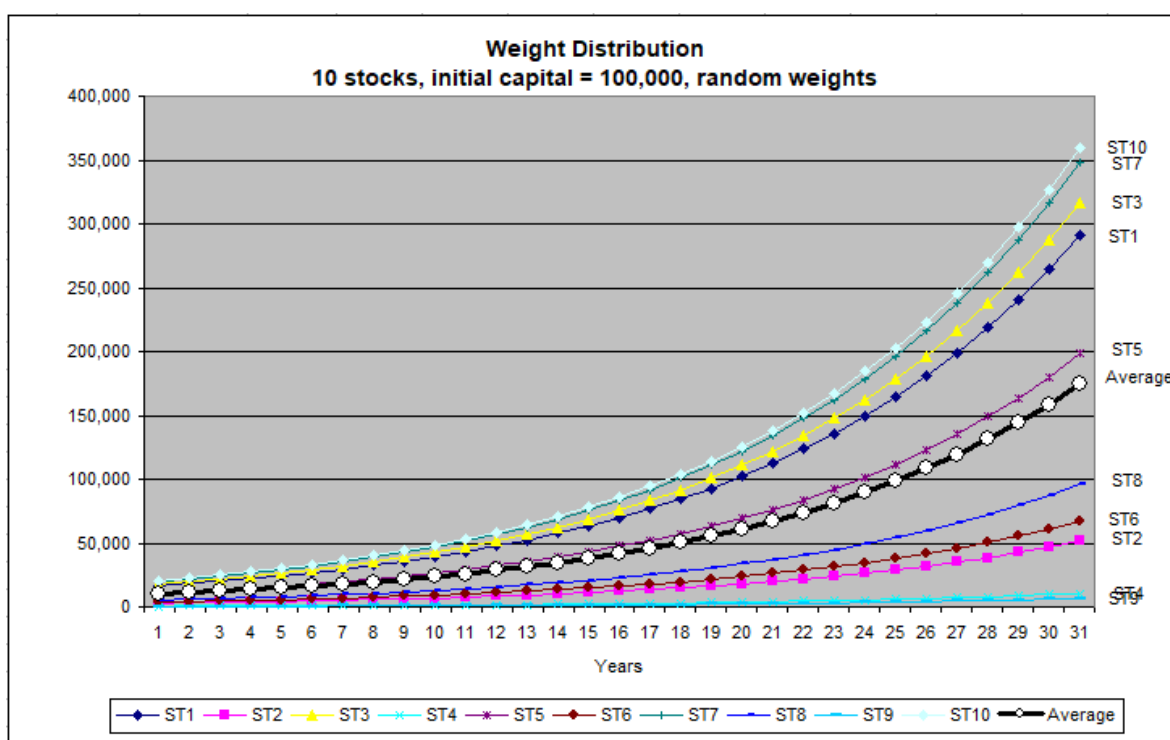
Showing the impact on trading is easy. Put 10 times more money on a position, you should get 1,227.5 shares. Flooring drops the 0.50 leaving 1,227 shares instead of 1,220 shares. This truncation will happen all the time on every trade, even partial trades. When you do tens of thousands of such trades, these small differences mount up since flooring is always in the same direction (reducing to the nearest whole number). A demonstration of this is illustrated by the program itself. With the only change being the initial capital, the 100k portfolio turned into 830k over the interval while the 1 million scenario increased to 10.0 million. So, technically, peanuts do count.

Since we are rebalancing the QQQ's 100 constituent stocks weekly, we have put some constraints on trade executability since all trades are executed on schedule as market orders. Even on partial trades, $\lfloor q_{i,d,j} \rfloor$ needs to be greater than 1 due to flooring, otherwise no trade. Therefore, it is required that the price change be sufficient to allow the buying or

selling of at least one share.

With the sum of weights equal to 1.0, ($\sum^J w_j = 1.0$), the composition of these weights might not matter much if all the stocks grow at the same rate. We can use the average rate of growth as applicable to all without loss of generalities. We would get the same end results. The chart below illustrates this:

Weight Distribution



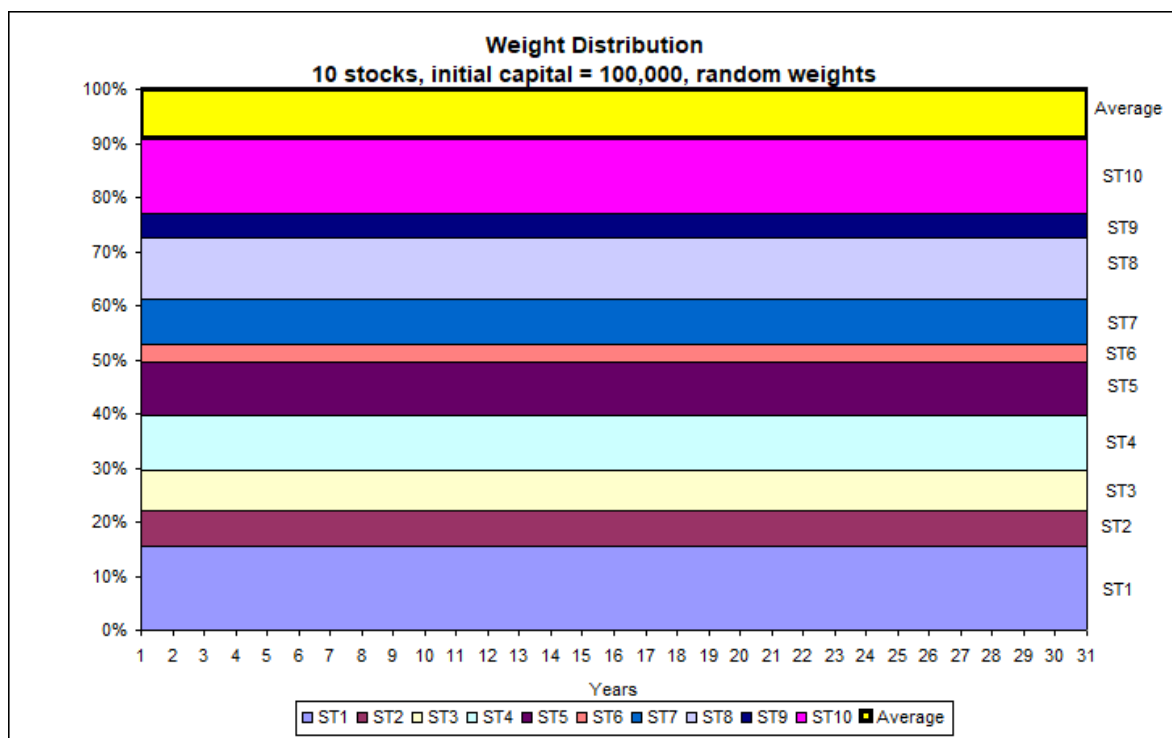
Each weight was randomly generated, then normalized so that their sum would total 1.0. We could run the test as many times as we wanted and the general average would not change, nor would the total outcome. Due to the random allocation of weights, we could not know which stocks would perform the most or in which order the stocks would come in.

Whatever the weights were, as long as they added to 1.0, the average growth rate (g) would be the same for all as in: $F(t) = F_0 \cdot (1 + g)^t$. You made a new test run and all the numbers for each stock would change, but, the total $F(t)$ and the average return g would remain the same.

On each rebalance, weights would return to their respective QQQ weights: $w_{i_{d,j}}$.

It is clear from the chart below (which displays the same data as the previous chart) that all weights, although randomly selected, remained constant over the 12.16-year trading interval.

Initial Normalized Random Weights



We have many strategies that use rebalancing. In this pure rebalancing play, we have a hard time showing that it was even beneficial, as shown in the previous section. Just holding QQQ for the duration would have produced a CAGR of 21.37% with no other effort than just taking the bet and holding on for those 12.16 years.

But we trade, nonetheless, with we hope, some other added advantages. Some of which could be better trade timing, moving in and out as the trend changes, better stock selection procedures, and better forecasting abilities than average. This also invites better protective measures and better trend definitions as well as better methods of play in order to show

some alpha: $F(t) = F_0 \cdot (1 + r_m + \alpha)^t$ since rebalancing alone did not seem to provide any. One could have bought QQQ and done better.

The **IN & OUT** strategy, referenced in related articles, displayed some of the above traits. In my simulated version, the top 100 momentum stocks were used and rebalanced daily which considerably increased the number of partial trades. It also increased performance. The flusher part of that strategy also had interesting side effects.

In all, it puts some emphasis on a rather mundane constant in a world of random-like behaviors: the position weight. It was shown to have an impact. Here, this strategy puts more emphasis on the highest valued stocks instead of using equal weights.

The number of stocks we trade is an administrative decision and in many cases remains constant over the entire trading interval. It can be used for risk diversification as well as self-controlled trade management. The above equation fixed the quantity to be traded for every trade taken. That it be over a simulation or some future undetermined trade just by setting the portfolio's position size proportional to the stock's weight in the portfolio.

We control the number of stocks (J) our strategies will trade. We set with how much capital our strategies will start with (F_0). We even determine for how long (t) our strategy will be applied. All that is left is to figure out: $g = r_m + \alpha$. Especially alpha which is where you will make a difference since average market performance $\bar{r}_m$ seems to be available to anyone.

Oct. 31, 2021

A Trading Strategy of Interest - Part III

or better yet

Use QQQ, Make the Money and Keep IT

However you want to trade stocks, the objective is to extract money from the process and not give it back.

It might not matter much how it is done as long as it is done (honestly and safely, evidently).

The methods used will depend on your knowledge and understanding of the game you intend to play, your trading capital, and your trading skills. This free trading strategy ([Creating your own index fund](#)) is the same as used in Part I and II. As said, I found the strategy interesting for the simplicity of its stock selection process, its pure-rebalancing play, and its overall performance.

Rebalancing every week at a scheduled time is an administrative decision requiring a single line of code (case in point: see program line 18). The strategy does not ask for your opinion, your forecast, how stocks are doing. Not even how you feel. Nonetheless, rebalancing does its job and it does have consequences and limitations.

Any trading strategy using some form of rebalancing will inherit some of the traits discussed here. For example, it is common practice to floor the number of shares purchased in order to avoid exceeding assigned weights which could otherwise lead to added margin fees.

Rebalancing could execute a lot of partial trades (here 60,000⁺) when making these inventory adjustments to return portfolio weights to their preset or newly determined values.

The Stock Selection

QQQ is composed of the top 100 highest weighted by market cap stocks on NASDAQ. They got there because they prospered and got big. They stay there because they get bigger. Those that flatter at the task of staying up there are replaced. They really are money making-machines, representative of the whole US economy.

In a single ETF, you have stealth portfolio diversification. These 100 companies have invested in their respective futures. They have thousands and thousands of employees, real estate holdings, products and services to sell, bonds, debts, proprietary stuff, and a lot more.

We have, in QQQ, an all-in-one package, the same bet as Mr. Buffett's bet on America. The stock selection process has been delegated to the issuer and manager of QQQ. And your input is not required or even solicited.

What you know is that QQQ has the top 100 richest companies listed on NASDAQ. And that is also a group of stocks you should follow, the stocks that tend to rise the most. You are in this game for the money, not to stagnate around, you have other things to do. You want stocks that are mostly moving up, not down. Even though there is a play there too.

You could buy QQQ outright and hold it for the duration, or trade its components as with this strategy hoping for a better return which might not materialize, but where you could certainly approach QQQ's own overall long-term performance level.

Your portfolio would be tracking a tracker which is tracking a market index. Therefore, almost evidently, you should get close to the same market index results.

In the previous section, weekly rebalancing was shown as almost mimicking QQQ's overall performance. The point was made that truncating the decimal part on the number of shares to be traded had

an impact on overall performance. The impact was made more visible when using smaller initial capital, but nevertheless, it applies all over.

QQQ's composition will change with time. Stocks failing to maintain their top 100 status are replaced with newcomers. We should expect that most of those changes occur near the bottom of this 100 list. This, at times, could generate more than 100 trades in a single week due to adding in the replacements.

As a trader or investor, your expectation is that those top 100 stocks will continue to grow, thus deriving your own bet on America.

Is QQQ risky? For one thing, it will not go down to zero overnight since it would require that all of its 100 constituent stocks go bankrupt. You will not see that next week either, almost surely... Can a component go bankrupt? For sure, but it will not happen while it is part of QQQ since it will lose its top 100 status and be liquidated and replaced long before that ever happens. Does QQQ only go up? Evidently no, but, over the long haul, it will fluctuate on its way up, even if at times, considerably. Explicitly saying there will be drawdowns, and that is more than almost surely...

The Rebalancing

The scheduled rebalancing is like mostly trading over a core position. Any price move that changed a stock's weighing is "corrected", forcing a partial trade of at least one share if the move is greater than $1/q_i\%$ as demonstrated in Part II.

It was shown that adding more capital generated more profits than expected. The reason was simple, with more money, more partial trades could qualify. The one share threshold was easier to reach with increasing position size since $1/q_i\%$ was getting smaller percentage-wise as q_i increased. Buying 10 shares of a 100-dollar stock will require at least a 10% move to buy another share, a 10\$ move. Buying 100 shares would only require a 1% price move, or only going up by 1\$ to add another share.

It was shown that the strategy turned 100k into 830k generating a 19.01% CAGR over its 12.24-year simulation interval (latest simulations). While increasing the initial stake to 1 million generated 10.0 million with a 20.6% CAGR. Something better than most long-term market averages.

The performance difference was mostly attributed to the truncation of the decimal part on the number of shares to be traded. Nothing was required except providing the initial cash, and monitoring the program while it ran (on schedule once a week).

Other Tests

I had to do more tests which you can do too. Copy the program, change its initial cash and run it freely on the QuantConnect website. It will give you the same results with all the details and nuances expressed here. Proving to yourself (the only person where it really matters) that it all holds together. You will find that there is simple math underneath it all.

A simulation tries to answer the question: *if I had done this or that in the past, what would have been the outcome?* Once a simulation has been performed, you immediately get the next most important question: *can this thing continue going forward?* Hence, your own bet on America.

Your Own Index Fund

Simulations from: Creating Our Own Index Fund											
Using QQQ's 100 constituent stocks											
		Start Date:	7/27/2009	(Rebalanced Weekly at 10:01 am)							
		End Date:	10/18/2021	Average			Extended to	Cost of ***			
		Liquidation	Total	Trades			20 Years	Recorded	Hit		
		Value	Trades	per week	CAGR **		Using CAGR	Drawdown	Rate	x_bar *	Leverage (%)
Leverage	Test	Initial Cap.									
1.00	#3	10,000,000	103,044,609	63,800	100.0	20.99%	451,621,035	28.00%	74%	1,615.12	0.00%
1.00	#2	1,000,000	10,070,521	56,913	89.2	20.76%	43,497,279	27.90%	71%	176.95	0.00%
1.00	#1	100,000	870,273	28,972	45.4	19.33%	3,427,047	27.50%	64%	30.04	0.00%
1.00	#8	50,000	400,402	19,382	30.4	18.52%	1,494,547	27.00%	60%	20.66	0.00%
1.30	#4	10,000,000	191,839,630	64,365	100.9	27.28%	1,245,880,240	35.20%	72%	2,980.50	-1.50%
1.50	#5	10,000,000	284,341,169	64,483	101.1	31.44%	2,369,283,832	39.70%	72%	4,409.55	-2.50%
1.75	#6	10,000,000	454,170,768	64,611	101.3	36.57%	5,090,825,150	45.10%	71%	7,029.31	-3.75%
1.95	#7	10,000,000	617,417,097	64,588	101.2	40.03%	8,406,357,599	51.10%	70%	9,559.32	-4.75%
* x_bar: average net profit per trade											
** CAGR: Compounded annual growth rate over 12.24 years											
*** Cost of leveraging (at 5%) will reduce CAGR by percentage shown											

The table above shows the outcome of such tests. In the top panel, only the initial capital was changed, nothing else, while in the bottom panel, only the impact of leverage was considered. What should we take out of these results?

Test #1 served as the basis with its 100k start. It is the one described in prior sections (here with a few days added). It managed a 19.33% CAGR making 28,972 trades with an average net profit per trade of \$30.04.

Test #2 raised the ante to 1 million. It generated a 20.75% CAGR doing 56,913 trades with an average net profit per trade of \$176.95. Increasing the position size increased the number of partial trades by 27,941 showing that the truncation is not such a trivial problem.

Test #3 raised the bar even higher with its 10 million initial stake. It generated a 20.99% CAGR doing 63,800 trades with an average profit per trade of \$1,615.12. We only have 100 stocks that can rebalance on a weekly basis. Over the trading interval, if all stocks participated every week, that would be 63,800 potential trades (638 weeks times 100). Therefore, with a 10 million initial capital, the strategy is reaching for its full potential in its expected number of trades. Is this understandable? Yes.

The Math

You put more on the line, your portfolio outcome should be proportional. The payoff matrix equation is the same. $F(t) = F_0 + \sum(H \cdot \Delta P)$.

You put up 10 times more money, it will multiply the quantities traded by 10. It is that simple: $10 \cdot F(t) = 10 \cdot [F_0 + \sum(H \cdot \Delta P)]$.

Prices will not change, they are the same for all. Moreover, all 100 stocks are scheduled for trading at 10:01 am once a week, making all exits and entries at the then-current trading price.

Using QQQ, the same 100 stocks will be traded for either portfolios: $10 \cdot F(t) = 10 \cdot F_0 + \sum(10 \cdot H \cdot \Delta P)$. Where you traded 10 shares you would now trade 100 and instead of requiring a 10% price move to purchase

another share, you would now only need a 1% move to do the same. The difference in profit, from expected, must come from somewhere. Over the short time span of a week, a 1% price move is more likely than a 10% move, thus the increase in partial trades.

Future Ballpark Expectations

Extending the strategy to 20 years using the same CAGR achieved over the past 12.24 years can provide an approximation, a ballpark figure, on the strategy's future expected performance. Comparing test #3 to test #1, where only the initial capital is considered shows a \$ 448 million difference in expected outcomes. A direct consequence of not raising more initial capital. We could view it as some opportunity cost. It was not a strategy request but an administrative decision that limited the capital at play. This also gives a value to not going for the added capital.

As the initial capital increased, the number of trades executed also increased where only the bet size should have increased. Usually, increasing the bet size generates about the same number of trades but with a higher average profit per trade, as should be expected: 10 times more on the table, close to 10 times more in profits, not more.

One explanation for the increase in the number of trades is this decimal truncation, the flooring of the number of shares that will be purchased or sold. As you increase the bet size, more price variations qualify for the minimum 1 share trade requirement. Whereas with rounding the number of shares, things might probably have evened out, but at times going slightly over full exposure which could trigger the use of margin and associated fees.

With the higher initial capital, we see an increase in the number of trades, an increase in CAGR, and an increase in the average net profit per trade (see $\bar{x}$, the $\bar{x}$ column). This, mostly due to the flooring of the number of shares traded. The differences are significant.

You have a strategy that does not outperform its tracker index, but, nonetheless, did beat long-term market averages over its past 12 years.

The stock selection is more than relatively secured, in fact, it is a reasonable bet on America. The stuff on which you can build retirement funds or grow your portfolio for whatever purpose. QQQ did more than outperform putting one's money in a bank account earning low interest.

How could you use this strategy? It got close to QQQ's overall return. Nonetheless, you could have done a little better just by buying QQQ outright. A single operation with little effort if any. No need for a computer, trading software, or automation. All that was required was taking a position and simply sitting it out.

Did you have more risk in putting 100k or 10 million on the table? Based on the above table, the max drawdown would have gone from -27.5% to -28.0%. We cannot say the difference was significant. Therefore, the very first consideration should be to find ways to get that 10 million, and even more, as initial capital. The effort would be well rewarded with not much-added risk.

Note that the hit rate in the first panel goes from 60% to 74% as you increase the initial capital. Also rising is the average net profit per trade $\bar{x}$ and the overall CAGR. Those 3 measures should have declined.

The program cannot improve by itself, especially, using rebalancing. And yet, the only change to the program was increasing initial capital, nothing else. Where is the Law of diminishing returns in this case? Does it kick in later?

Leveraging

The bottom panel in the above table shows the impact of applying some leverage. Evidently, there is a cost associated with it. However, these cost will only be on the leveraged part of the portfolio:

$$F(t) \cdot (1 + Lev) = (1 + Lev) \cdot (F_0 + \sum (H \cdot \Delta P)) - exp$$

For example, a 30% leveraged account would have leveraging fees on only 30% of the portfolio: $0.30 \cdot F(t)$. Leveraging would also raise trading capital by 30%: $F(t) \cdot (1 + 0.30)$. This will not have that much of an impact on the number of trades since we will be reaching what appears

as a slowly approaching upside limit offered by the inherent mechanics of rebalancing. You have only 100 stocks that can rebalance on a weekly basis. Plus, once in a while, a few more, due to replacements.

Evidently, as we increase leveraging, drawdowns increase.

As compensation, we do see an increase in CAGR that exceeds the added leveraging expenses. For instance, at 30% leverage, the estimated cost with a 5% leveraging fee would be: $0.30 \cdot F(t) \cdot 0.05$. This would reduce overall CAGR by -1.5%, but nonetheless increase CAGR from 20.99% to 27.28%. Thereby, increasing CAGR, net of leveraging expenses, by 6.3%, which is 4.2 times better than its cost.

With leverage at the 95% level, the CAGR will be reduced by -4.75% ($0.95 \cdot 0.05 = 0.0475$). Looking at the CAGR for those leveraged scenarios, there is more than enough to cover those added fees even if they will be considerable. Leveraging just becomes another tool, and also, an added cost of doing business.

So, just based on the above table, in the first panel, you gain a higher CAGR by putting more on the table. As you put more on the table, the number of trades increased toward its estimated full potential of 63,800 trades.

Putting less on the table is like underplaying one's hand.

The money, (a.k.a.) the profits, were there for the taking, but we lacked the capital to take advantage of it. And it is one of the reasons why you did not get more.

Whereas, in the bottom panel, the number of trades went up only a little (by about 1.1%, on average) but still remained near full potential, as should be expected. The slightly higher number of trades could be explained by the purchase of replacements for the dropped stocks in order to maintain the portfolio's 100 stock limit. Also, as leveraging increased, more partial trades could qualify.

It is only after the 75% leverage mark that the number of trades decreased, but nonetheless, managed to keep a 70% hit rate.

Due to the rebalance scheme, there is an upper limit on the potential number of trades per week. It is highly related to the number of stocks in the portfolio and their respective price variations.

You find that as the leverage increased, the drawdowns also increased. That should be expected, we increased the bet size by leveraging, thereby amplifying the impact of volatility.

Nonetheless, such a trading strategy could be used to complement others aimed at reducing overall portfolio volatility. There is a cost to leveraging, therefore, maybe, be the one providing the added funds and collecting those fees (they get to be considerable), lend the needed margin to your trading strategy.

Increasing your initial capital by the same percentage amount as the margin is another administrative move. It does not have the same effect. All it does is increase the initial capital while leveraging will also impact all trades and generate much more. Which level of performance do you want?

Raising Initial Cash – No Leveraging

Increasing Initial Capital - No Leverage											
		Start Date:	7/27/2009	(Rebalanced Weekly at 10:01 am)							
		End Date:	10/18/2021								
				Average		Extended to					Cost of
				Trades		20 Years		Recorded	Hit		Leverage
Leverage	Test	Initial Cap.	Liquidation	Total	per week	CAGR	Using CAGR	Drawdown	Rate	x_bar	(%)
			Value	Trades							
1.00	#9	13,000,000	135,083,674	64,113	100.5	21.04%	592,664,194	28.00%	74%	2,106.96	0.00%
1.00	#10	15,000,000	155,905,936	64,207	100.6	21.05%	684,069,318	28.00%	74%	2,428.18	0.00%
1.00	#11	17,500,000	181,929,727	64,321	100.8	21.05%	798,344,640	28.00%	74%	2,828.47	0.00%
1.00	#12	19,500,000	202,747,266	64,404	100.9	21.05%	889,731,020	28.00%	74%	3,148.05	0.00%

Note that all 4 of the tests above maintained a hit rate of 74% and a drawdown of -28%, the same as in test #3. As you increased the initial capital, the CAGR also stayed about the same near 21.05%.

The difference between using leverage and only raising initial capital is considerable. The difference comes from what is being compounded. In

the above chart, the initial cash increase raised the portfolio's outcome close to the percent increase (test #9). 30% more cash, about 31% more in total liquidation value. While the 30% leverage (test #4) increased the liquidation value by 86.17%. Leveraging enabled continuous reinvestment of leveraged profits.

The 74% hit rate is not the result of some better market timing. In reality, this strategy is only reacting to market turbulence. We could say that it operates on the fumes of variance. If the price does not move by enough, that is more than $1/q$, there is no trade.

The strategy makes no other prediction than its initial assumption, its bet on America. Understandably, the majority of sales will be at a profit, most shares are sold on the way up. For instance, on test #9, some 74% of trades (47,463) out of 64,113 were positive.

Buying is done when prices decline. There are no immediate losses acknowledged on these "purchases". We just add to positions without liquidating (except for the stocks falling off the top 100 list). The 74% hit rate is simply due to the structure of the designed game which strictly specified the rules of engagement. It is the result of the strategy's trade dynamics.

Stocks Traded – QQQ Weights

Assets Sales Volume																		
AAPL	MSFT					AMZN				FB		GOOG		INTC		CSCO		
	NVDA	CMCSA	TSLA	AMGN	NFLX	ADBE	PYPL	QCOM	GILD	PEP	AVGO	COST	TXN	SBUX	CHTR	PCLN		
	KFT	TMUS	BIIB	BIDU	AMAT	MU	WBA	ADP	CELG	EBAY	ESRX	KHC	RE...	N...	O...			

The above chart mimics the weights found in QQQ for each position. The 10 largest corporations account for the lion's share, 55% of the total. The other 90 stocks generated the remaining 45%. All the tests, from #1 to #16 showed almost identical distributions. The reason is simple. QQQ is a market-cap-weighted tracker. And therefore its weights will tend to match and follow closely its index counterpart.

It is now your move, make your own bet on America.

Could you make more? Yes, definitely, and for sure. You got a copy of the program, you can do all the same tests shown here and more. However, I would suggest adding some protection aimed at reducing volatility and drawdowns. Add some timing signals to improve profit margins. Add some gaming to enhance overall returns. Improve on the strategy's design.

It is always up to you to prove to yourself that you can do it.

A simulation is just to show it is possible and worthwhile. I can only encourage you to plan for your next 20 years.

Also, a portfolio does not need to retire at 65, it can live another 30⁺ years. You will find out, in the end, that it was all about you and your choices.

In the meantime, consider adding capital and applying leverage at the same time. It would result in something like in the table below.

Raising Initial Cash – And Using Leverage

Increasing Initial Capital - With Leverage											
		Start Date:	7/27/2009	(Rebalanced Weekly at 10:01 am)							
		End Date:	10/18/2021	Average		Extended to				Cost of	
			Liquidation	Total	Trades	20 Years		Recorded	Hit		
Leverage	Test	Initial Cap.	Value	Trades	per week	CAGR	Using CAGR	Drawdown	Rate	x_bar	Leverage (%)
1.30	#13	13,000,000	253,765,498	64,579	101.2	27.42%	1,654,349,253	35.20%	73%	3,929.54	-1.50%
1.50	#14	15,000,000	435,143,397	64,730	101.5	31.60%	3,642,562,808	39.70%	72%	6,722.44	-2.50%
1.75	#15	17,500,000	813,492,142	64,852	101.6	36.76%	9,168,185,734	45.10%	71%	12,543.83	-3.75%
1.95	#16	19,500,000	1,235,365,132	64,785	101.5	40.26%	16,927,290,505	51.10%	70%	19,068.69	-4.75%

Both leverage and added capital are considered. Notice the range of the outcomes, check test #8 with its 50k initial stake to test #16.

All done using the same strategy, the same trading logic, the same rebalancing on the same stocks, on the same dates at the same times. Yet, you can go from \$ 1,494,547 to \$ 16,927,290,505 just by managing the strategy's available cash resources. It should give you some motivation to find more capital right from the start of your 20-year journey.

You know that if you add more leverage to this strategy, it will have larger drawdowns. Then, based on the level you want to reach, maybe it would be time to investigate trading methods that would reduce those drawdowns to more acceptable levels. However, note that the drawdowns in the above chart are the same as in the second panel of the first table above.

This says that it is not raising the initial capital that is raising the drawdowns, it is the leveraging. The CAGR on the second panel in the first table is about the same as in the table above.

However, this CAGR will be reduced by the cost of leveraging. But these costs are taken directly from the trading account as you progress. Meaning that it is not fresh capital that you have to pour in, those will be borrowed funds on which you will pay leveraging fees as you go. And your ongoing market profits will be able to pay for it and some.

All the tests presented were achieved using the stated initial capital and leverage. Using more cash, just as setting some acceptable leverage, are both administrative decisions. And since we are dealing with 100 stocks, the concentration risk factor has been reduced considerably.

You could run the program with any numbers you want. Change the initial capital to whatever amount, and set the leverage to what you might find acceptable, including zero. And if the numbers are within those presented, the program will comply and deliver something between test #1 and test #16.

The strategy can do test #1 just as it can do test #16. The difference is how you consider money and how you can get it. You can put more capital on the table. Good for you. You can tolerate having your program use some leverage, then, there is a reward attached to that too.

These tests, from #1 to #16, are all doable by anybody having the resources. It does not even require that you have any kind of trading skills. You do not need special training, and to top it off, the trading script is free.

You are afraid or convinced that at some point the markets will go down. Why should you let your program run? Shut it down, liquidate all positions, go to the sidelines. You can always restart the program later under the same ending conditions or set new initial conditions. It is not because you have a program, that you cannot stop it at will. You are dealing with a machine having a big red emergency button that you can press at any time. You are in charge. Protecting your portfolio should be a major part of your trading strategy.

You have, here, a trading strategy that has nothing in it: no secret sauce, no indicators to follow or guiding trade execution, no signals to trigger trades, no request for fundamental data. It is not the best but is pretty interesting considering.

You want more, you just put more in. You provide the input: the how much you put in it. It is all on you and ultimately how much you want it.

A 10 million initial account with 30% leverage can reach the billionaire status in 20 years. And if you did not want to be bothered by the programming (actually changing two numbers) and monitoring your account, simply buy QQQ outright, sit down, and relax for the next 20⁺ years. The simulations were to show that buying QQQ was not that bad an idea and trading this strategy could do about the same.

Sometimes you find problems that you think will require complex solutions because they are, in fact, quite complicated. And then, you find readymade solutions, designed by others, when you were not even looking their way.

Make your own bet on America. You cannot say you do not have the tools. They are there, free for the taking. If you can do better, do it. However, you should consider this strategy as the very minimum you could do. If you do less, the question should be: why? You had this available all along.

Finally, it is always your choice to do or not to do.

Sincere and best wishes.

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